



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 vide notification number F-9-12/2001-U-3 of the Government of India)
Re-Accredited by NAAC with 'A++' Grade

Karan Dave, 26070122507, CSE B3, 2025-2029

ASSIGNMENT No: 5

TITLE: Develop a Java application demonstrating the use of ArrayList, Vector, and StringBuffer for dynamic data storage, management, and string manipulation.

Problem Statement

Develop a Java application demonstrating the use of ArrayList, Vector, and StringBuffer for dynamic data storage, management, and string manipulation.

Learning Objectives

To implement dynamic data storage and string manipulation using ArrayList, Vector, and StringBuffer.

Theory

ArrayList and Vector are dynamic array implementations provided by the Java Collections Framework. ArrayList offers faster performance in single-threaded applications, whereas Vector provides synchronized operations for thread safety. StringBuffer is a mutable sequence of characters that allows efficient string modification without creating new objects. These classes are commonly used in applications requiring dynamic memory allocation and efficient string processing.



Exercise

1. Create a To-Do List application using ArrayList to store tasks and StringBuffer to display tasks.

Input-

```
*pehlojche.java X
1 package pehlojche;
2 import java.util.Scanner;
3 public class pehlojche {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         System.out.println("==== Subject To-Do List =====");
7         System.out.println("1. Mathematics");
8         System.out.println("2. Science");
9         System.out.println("3. English");
10        System.out.println("4. Computer");
11        System.out.println("5. Social Science");
12        System.out.print("Enter your choice (1-5): ");
13        int choice = sc.nextInt();
14        switch (choice) {
15            case 1:
16                System.out.println("Mathematics To-Do:");
17                System.out.println("- Solve Chapter");
18                System.out.println("- Complete Exercise");
19                System.out.println("- Revise formulas");
20                break;
21            case 2:
22                System.out.println("Science To-Do:");
23                System.out.println("- Read Chapter");
24                System.out.println("- Draw diagrams");
25                System.out.println("- Learn definitions");
```

```
26        break;
27        case 3:
28            System.out.println("English To-Do:");
29            System.out.println("- Read Lesson");
30            System.out.println("- Learn grammar");
31            System.out.println("- Write an essay");
32            break;
33        case 4:
34            System.out.println("Computer To-Do:");
35            System.out.println("- Practice Java");
36            System.out.println("- Complete lab work");
37            System.out.println("- Revise programming concepts");
38            break;
39        case 5:
40            System.out.println("Social Science To-Do:");
41            System.out.println("- Read History chapter");
42            System.out.println("- Learn important dates");
43            System.out.println("- Complete map work");
44            break;
45        default:
46            System.out.println("Invalid Choice!");
47        }
48        sc.close();
49    }
50 }
```



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 vide notification number F-9-12/2001-U-3 of the Government of India)
Re-Accredited by NAAC with 'A++' Grade

Output-

```
===== Subject To-Do List =====
```

1. Mathematics
2. Science
3. English
4. Computer
5. Social Science

Enter your choice (1-5): 1

Mathematics To-Do:

- Solve Chapter
- Complete Exercise
- Revise formulas

```
===== Subject To-Do List =====
```

1. Mathematics
2. Science
3. English
4. Computer
5. Social Science

Enter your choice (1-5): 2

Science To-Do:

- Read Chapter
- Draw diagrams
- Learn definitions

```
===== Subject To-Do List =====
```

1. Mathematics
2. Science
3. English
4. Computer
5. Social Science

Enter your choice (1-5): 3

English To-Do:

- Read Lesson
- Learn grammar
- Write an essay



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 vide notification number F-9-12/2001-U-3 of the Government of India)

Re-Accredited by NAAC with 'A++' Grade

===== Subject To-Do List =====

1. Mathematics

2. Science

3. English

4. Computer

5. Social Science

Enter your choice (1-5): 4

Computer To-Do:

- Practice Java

- Complete lab work

- Revise programming concepts

===== Subject To-Do List =====

1. Mathematics

2. Science

3. English

4. Computer

5. Social Science

Enter your choice (1-5): 5

Social Science To-Do:

- Read History chapter

- Learn important dates

- Complete map work



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 vide notification number F-9-12/2001-U-3 of the Government of India)
Re-Accredited by NAAC with 'A++' Grade

2. Create a Student Course Registration System using ArrayList to store the list of courses registered by a student and StringBuffer to generate and display the registered course list. The application should allow users to add, remove, and view registered courses.

Input-

```
*StudentCourseRegistration.java
1 package pehlojche;
2 import java.util.ArrayList;
3 import java.util.Scanner;
4 public class StudentCourseRegistration
5 {
6     public static void main(String[] args)
7     {
8         Scanner sc = new Scanner(System.in);
9         ArrayList<String> courses = new ArrayList<String>();
10        int choice;
11        do
12        {
13            System.out.println("\n==== Student Course Registration System =====");
14            System.out.println("1. Add Course");
15            System.out.println("2. Remove Course");
16            System.out.println("3. View Registered Courses");
17            System.out.println("4. Exit");
18            System.out.print("Enter your choice: ");
19            choice = sc.nextInt();
20            sc.nextLine();
21            switch (choice)
```

```
22        {
23            case 1:
24            {
25                System.out.print("Enter course name: ");
26                String course = sc.nextLine();
27                courses.add(course);
28                System.out.println("Course added successfully.");
29                break;
30            }
31            case 2:
32            {
33                System.out.print("Enter course name to remove: ");
34                String course = sc.nextLine();
35                if (courses.remove(course))
36                {
37                    System.out.println("Course removed successfully.");
38                }
39                else
40                {
41                    System.out.println("Course not found.");
42                }
43                break;
44            }
45        }
46    }
47 }
```



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 vide notification number F-9-12/2001-U-3 of the Government of India)
Re-Accredited by NAAC with 'A++' Grade

```
43         break;
44     }
45     case 3:
46     {
47         if (courses.isEmpty())
48         {
49             System.out.println("No courses registered.");
50         }
51         else
52         {
53             StringBuffer sb = new StringBuffer();
54             sb.append("\nRegistered Courses:\n");
55             for (String course : courses)
56             {
57                 sb.append("- ").append(course).append("\n");
58             }
59             System.out.println(sb.toString());
60         }
61     }
62     break;
```

Problems Javadoc Declaration Console Properties Sponsor the Eclipse IDE
StudentCourseRegistration [Java Application] C:\Users\pdave\p2\pool\plugins\org.eclipse.justi.openjdk.hotspot.jre.full.win32.x86_64_21.0.11.v20260515-1531\jre\bin\javaw.exe (30-Jul-2026, 12:09:33 pm elapsed 0:03:17) [pid: 8872]
View Registered Courses Writable Smart Insert 79: 1: 2674
ENG IN 46% 12:12 30-07-2026

```
63     }
64     case 4:
65     {
66         System.out.println("Thank You!");
67         break;
68     }
69     default:
70     {
71         System.out.println("Invalid Choice.");
72     }
73 }
74 }
75 while (choice != 4);
76 sc.close();
77 }
78 }
79 }
```

Problems Javadoc Declaration Console Properties Sponsor the Eclipse IDE
StudentCourseRegistration [Java Application] C:\Users\pdave\p2\pool\plugins\org.eclipse.justi.openjdk.hotspot.jre.full.win32.x86_64_21.0.11.v20260515-1531\jre\bin\javaw.exe (30-Jul-2026, 12:09:33 pm elapsed 0:03:23) [pid: 8872]
View Registered Courses Writable Smart Insert 79: 1: 2674
ENG IN 46% 12:12 30-07-2026



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 vide notification number F-9-12/2001-U-3 of the Government of India)
Re-Accredited by NAAC with 'A++' Grade

Output-

```
===== Student Course Registration System =====
```

1. Add Course
2. Remove Course
3. View Registered Courses
4. Exit

```
Enter your choice: 1
```

```
Enter course name: java
```

```
Course added successfully.
```

```
===== Student Course Registration System =====
```

1. Add Course
2. Remove Course
3. View Registered Courses
4. Exit

```
Enter your choice: 1
```

```
Enter course name: python
```

```
Course added successfully.
```

```
===== Student Course Registration System =====
```

1. Add Course
2. Remove Course
3. View Registered Courses
4. Exit

```
Enter your choice: 3
```

```
Registered Courses:
```

- java
- python



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 vide notification number F-9-12/2001-U-3 of the Government of India)
Re-Accredited by NAAC with 'A++' Grade

```
===== Student Course Registration System =====
```

```
1. Add Course
```

```
2. Remove Course
```

```
3. View Registered Courses
```

```
4. Exit
```

```
Enter your choice: 2
```

```
Enter course name to remove: python
```

```
Course removed successfully.
```

```
===== Student Course Registration System =====
```

```
1. Add Course
```

```
2. Remove Course
```

```
3. View Registered Courses
```

```
4. Exit
```

```
Enter your choice: 3
```

```
Registered Courses:
```

```
- java
```

```
===== Student Course Registration System =====
```

```
1. Add Course
```

```
2. Remove Course
```

```
3. View Registered Courses
```

```
4. Exit
```

```
Enter your choice: 4
```

```
Thank You!
```

```
|
```